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OBSERVING TILTS IN MIDCRUSTAL ROCKS BY PALEOMAGNETISM: EXAMPLES FROM SOUTHEAST BRITISH COLUMBIA

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Abstract. Metamorphic core complexes that originated in the middle crust have been exposed by uplift and extension. Indicators of paleohorizontal planes are few or absent, thus it is difficult to determine the degree of tilting that might have accompanied this process. However, if aberrant paleomagnetic directions (paleodirections) are observed in core complexes or their associated intrusive rocks, and if large-scale rotation and translation can be excluded, then these paleodirections can be used to calculate the sense and magnitude of tilting. We describe paleodirections from mid-crustal rocks in southeastern British Columbia, where uplift and extension occurred in Eocene time about a central culmination aligned north-south. Several large extensional faults dip both to the west and to the east of this central culmination. The Eocene Granby pluton (49.3°N,118.5°W), situated to the west of the main culmination, has a mean direction of magnetization ($D°$, $I°$) 309°,41° ($\alpha_{63}°$ = 7° the standard error) and paleopole at 44°N,138°E ($A_{63}°$ = 7°). To the east of the main culmination, a 20 km long section of altered Eocene Syringa dikes and associated metadiorite (49.3°N, 117.7°W) has a mean direction of 56°,47° ($\alpha_{63}°$ = 3°) and paleopole at 56°N,319°E ($A_{63}°$ = 3°). The paleopoles are very different from one another, and from those calculated from contemporaneous (53 to 49 Ma) rocks of the North American craton to the east. They also differ from those observed from coeval bedded volcanic rocks to the west of the sampling area. Hence, these large aberrancies cannot be caused by translations. We suggest that they are caused by tilting which occurred during the extension process. The tilts are in opposite senses on either side of the central culmination. They are estimated to be down 38° ± 8° at 99°E for the Granby Pluton, and down 41° ± 4° at 270°W for the dikes and metadiorite. These tilts are consistent with the Wernicke model for simple extensional shear.

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INTRODUCTION

Rocks of the deep crust, whether of metamorphic or plutonic origin, acquire a magnetization in the direction of the geomagnetic field (paleodirection) as they are uplifted and cooled [Pullaiah et al., 1975]. Such "uplift" magnetizations are amongst the simplest forms of remanent magnetization. They have been extensively observed in Precambrian rocks, especially in the Grenville Province of eastern Canada [Berger et al., 1979; Irving et al., 1974]. For rocks containing fine-grained magnetite, 580°C is the critical magnetic blocking temperature, although if cooling is slow, the blocking temperature may be lowered, but not by more than about 20°C [York, 1978]. If uplift is rapid, say 1 mm yr^{-1}, then a crustal layer 10 km thick would acquire an essentially uniform paleodirection, because significant apparent polar wander rarely occurs in an interval as short as 10 Ma. Tilting following uplift and cooling would be recorded as aberrant paleodirections, which can be used to estimate the attitude of the paleohorizontal plane just prior to uplift.

Paleomagnetism has long been used in this general fashion as a means of estimating tilt in rocks of deep-seated origin [e.g. Hood, 1961; Shaver and McWilliams, 1987], and in this paper we describe examples as a contribution to the study of the kinematics of the Eocene extensional zone of southeastern British Columbia.

CAUSES OF ABERRANCY

Aberrant paleodirections may have causes other than tilting, and only when these other causes can be eliminated are tilting interpretations permissible. The general

principles are set out in Figure 1. Inclinations that are shallower than those expected may be caused either by tilting or by translation of the sampling locality from the south if in the northern hemisphere, or by translation from the north if in the southern hemisphere (Figure 1a). Aberrant declinations may be caused either by tilting or by rotation of the sampling locality about a local vertical axis (Figure 1b). If for other reasons large-scale translation and rotation can be shown not to have occurred, then tilting is the most likely cause, and its direction and magnitude may be calculated by rotating the aberrant paleodirection along a small circle into the reference direction (the rotation T of Figure 1b). Such a reconstruction is correct only if tilting occurred about an axis which is close to horizontal.

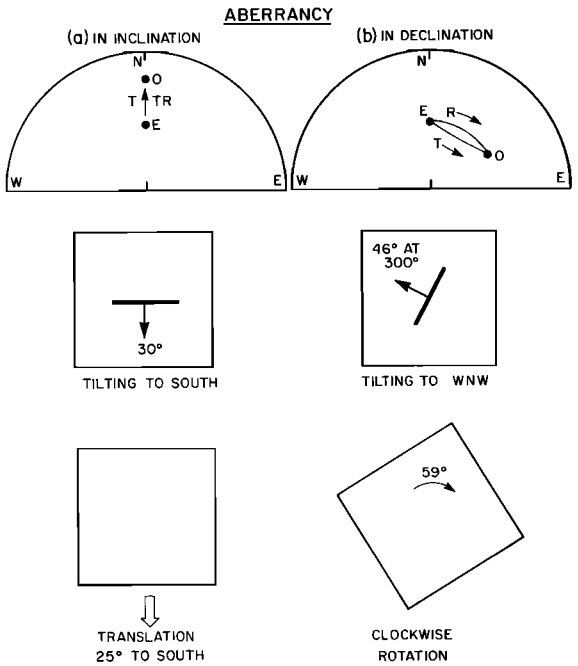


Fig. 1. Causes of aberrant paleodirection in a zone of tectonic disturbance. E is the expected paleodirection calculated from the adjacent craton, O is the paleodirection observed at the test locality in the zone of disturbance. Aberrancy in inclination (a) or declination (b) can be caused either by tilting, or by translation and rotation about a local Euler pole. The arrows labelled T, TR and R show the effect that tilting, translation and rotation respectively have on the paleodirection.

EXTENSION AND MAGNETIZATION

In most extensional systems the main faults are normal and dip gently. An important feature is that across these faults lower-grade rocks are superimposed on higher-grade metamorphic rocks (core complexes). The hanging wall is divided into a series of blocks (e.g., in the region of the Okanagan Valley Fault, as shown by Bardoux [1985] and Tempelman-Kluit and Parkinson [1986]).

To explain such a pattern, a simple shear extensional model has been proposed by Wernicke [1985] and discussed by Buck [1988]. In Figure 2 a hypothetical magnetization process which makes use of this model is proposed. Initially the upper crust will have magnetizations acquired by many processes during its earlier history.

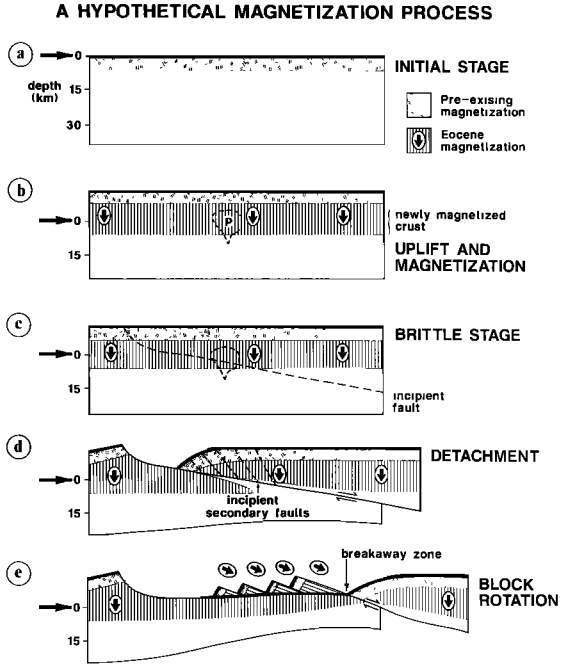


Fig. 2. Possible magnetization and tilting process for rocks originally at midcrust based on ideas of Wernicke [1985] and Buck [1988]: (a) initial stage; (b) uplift of the area, followed by cooling through magnetic blocking temperatures and acquisition of magnetization, "Coryell" plutons (P) may have intruded at this stage; (c) breaking of an extensional fault, as a thick section of the crust becomes brittle; (d) sliding of the hanging wall along the fault, causing the occurrence of secondary faults and the rotation of blocks close to the breakaway zone; (e) the hanging-wall blocks and the paleodirections acquired at stage (b) are now rotated.

During initial uplift, plutons are intruded and volcanics extruded at higher levels. The crust cools and rocks located at mid-crustal depths acquire a thermoremanent magnetization as they pass through their blocking temperatures (Figure 2b). A colder crust is more brittle, and failure is then initiated by the formation of a large, steeply-dipping normal fault cutting through the lithosphere (Figure 2c). Hanging wall rocks slide down along this fault. Flexural response of the hanging wall causes rotation of the primary fault at shallow depths. Then a new (secondary) fault is formed in the upper levels at an angle dependent upon the strength of the crust and the magnitude of the stresses applied. As rifting proceeds,

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a series of comparatively shallow secondary faults are produced (Figure 2d). In this model, the rotation of secondary shallow faults and the magnetized blocks they bound ceases as the locus of the breakaway zone recedes (Figure 2e). An important assumption of the above model is that uplift and acquisition of magnetization precede the generation and tilting of secondary faults. If this assumption is incorrect and magnetization occurred after tilting, then aberrant paleodirections would not be observed. The actual observation of aberrant paleodirections would imply that this assumption is entirely or partially correct.

Wernicke's model was developed for the Basin and Range region, where supracrustal bedded volcanic and sedimentary rocks are common and geological dips can be measured. However, in the central culmination zone of the extension zone of SE British Columbia, supracrustal rocks have been removed either by erosion or by tectonic denudation, exposing the middle crust. These mid-crustal rocks contain no visible record of the attitude of the paleohorizontal plane immediately prior to tilting. Hence, it is not known whether they too are tilted in the manner observed elsewhere in the shallower bedded supracrustal sequences.

APPLICATION TO SOUTHEAST BRITISH COLUMBIA

The study region is in the Omineca Belt in the eastern metamorphic core zone of the Cordilleran orogen (Figure 3). During Eocene time, widespread uplift and extension occurred (see Armstrong [1982] and references therein). A model describing the timing and manner of this extension which is estimated to be approximately 25% has been described recently by Parrish et al. [1988].

The crust fractured as it was uplifted, and this fracturing took the form of large low-angle normal faults along which extension occured by slippage (Figure 4). Generally these extensional faults trend north-south perpendicular to the axis of extension and dip outward on either side of a

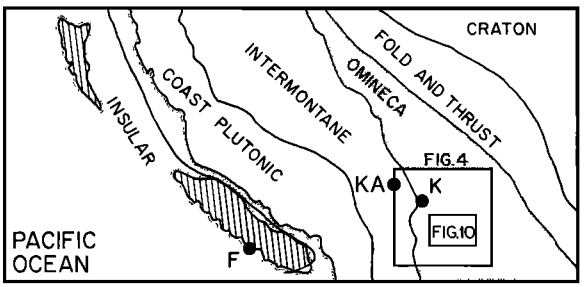


Fig. 3. Tectonic belts of southern British Columbia and the localities discussed in this paper. The larger box is the approximate region of Eocene extension shown in Figure 4. The smaller box, labelled Figure 10, is the sampling area. K (Kelowna Volcanics) KA (Kamloops Volcanics) and F (Flores Volcanics) are the locations at which studies have been made of Eocene volcanic rocks elsewhere in British Columbia (see Table 4).

central culmination, the Kettle-Grand Forks core complex (KC of Figure 4). Extensive U-Pb zircon dating indicates that the easterly dipping faults were active between 58 to 52 ago whereas westerly dipping faults were active between 52 and 48 Ma [Parrish et al., 1988]. The faults depicted by Parrish et al. [1988] are either planar as in the case of the Slocan Lake Fault, or slightly curved. This model of extension is different from the simple shear model of Wernicke in that the blocks between the major faults slide downwards relative to each other with minimal strain within blocks (compare Figures 2 and 4).

Recently Bardoux and Irving [1989] have described results of a pilot study undertaken in a zone of Eocene extension near Castlegar (British Columbia), which indicate that post-uplift tilting of as much as 40° has occurred. This implies that tilts of the magnitude encountered in supracrustal sequences of extensional zones extend also to deep-seated plutons and metamorphic rocks of the middle crust. The purpose of our study was to determine if such tilts occurred more generally in the extensional zone of southeast British Columbia, and particularly if they occurred symmetrically about the central culmination of this zone of extension.

MAGNETIZATION

Measurements were made on a Schonstedt spinner magnetometer (SSM2) controlled by an IBM/PC. Pilot specimens (at least two per site) were demagnetized thermally and by alternating magnetic fields (AF). Thermal demagnetization was carried out using a Schonstedt TSD-1 furnace, and AF demagnetization using a Schonstedt GSD-5 tumbling demagnetizer.

Magnetizations are of three types, designated A, B, and C. A-type is shown in Figure 5. A magnetically soft overprint is removed by treatment in a 10 mT AF. The natural remanent magnetization (NRM) is directed approximately parallel to the present earth's field and is interpreted as a viscous remanent magnetization (VRM) of recent origin. The principal magnetization is well-defined by linear decay to the origin of orthogonal plots (Figure 5b), and by an end-point reached after cleaning in alternating fields of 10 mT and higher (Figure 5a). The unblocking temperatures are in the range 500 to 580°C. The direction of the principal component in A-type magnetization was obtained by choosing points centrally placed on the straight lines trajectories. The cleaning fields and temperatures used were 20 or 30 mT and 550°C.

AF demagnetization of B-type magnetization is shown in Figure 6. The orthogonal plot shows linear decay but not to the origin. The lines are less well-defined than in A-type. The direction of the principal component of B-type magnetizations was obtained by least-squares analysis, usually in the range 20 to 80 mT. It is shown by the star on Figure 6. A small magnetization with high coercive force appears also to be present. The origin of this small magnetization is not known, and neither AF nor thermal magnetization yielded satisfactory end-points. Stepwise thermal demagnetization gave unblocking temperatures distributed between 300 and 550°C.

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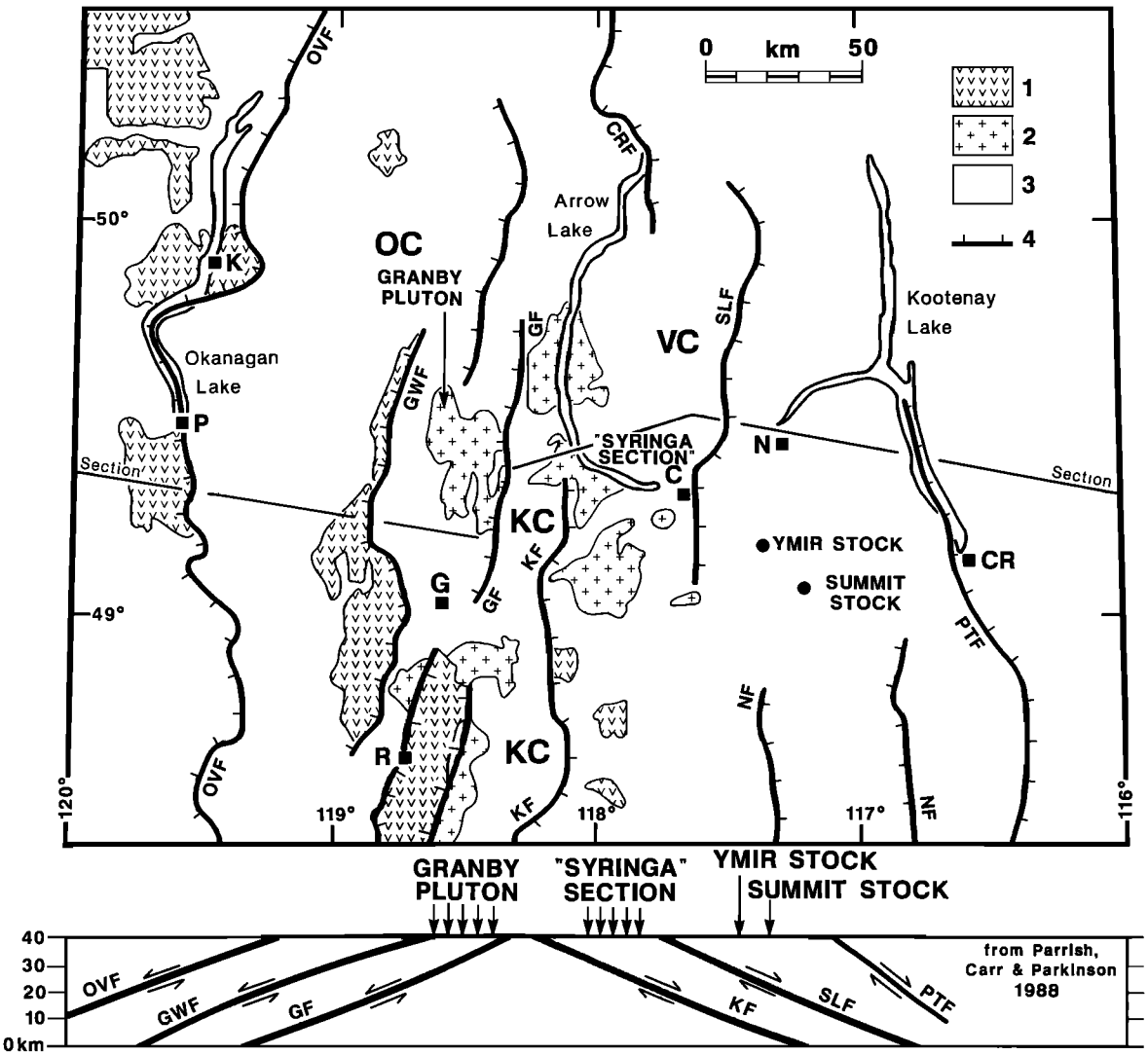


Fig. 4. Geological sketch-map of the Eocene extensional zone of southeastern British Columbia and northern Washington, compiled from Parrish et al. [1988]. Tilt estimates have been obtained from four localities, - Granby Pluton, "Syringa section", and Ymir and Summit stocks. The hypothetical section below is drawn at a somewhat smaller horizontal scale, but with vertical and horizontal scales the same. The low-angle extensional faults shown are the Okanagan Valley fault (OVF), the Greenwood fault system (GWF), the Granby River Fault (GF), the Kettle River fault (KF), the Columbia River fault (CRF), the Slocan Lake fault (SLF), the Newport fault (NF) and the Purcell Trench fault (PTF). Ticks are on the downthrow side. The Valkyr shear zone predates the events discussed herein and has been omitted. Major core-complexes are, Kettle-Grand Forks (KC), Okanagan (OC), Valhalla (VC). 1, Eocene syn-tectonic volcanics. 2, Eocene syn-tectonic plutons. 3, older rocks. Individual sampling sites are given in Figure 10. The towns of Castlegar (C), Creston (CR), Grand Forks (G), Kelowna (K), Nelson (N), Penticton (P), and Rossland (R) are marked.

Several samples had incoherent magnetizations that could not be isolated by demagnetization, and no interpretable information was obtained. These are C-type magnetizations.

GRANBY PLUTON

The Granby pluton is a large syenite-diorite body located 15 km north of Grand Forks. It belongs to the

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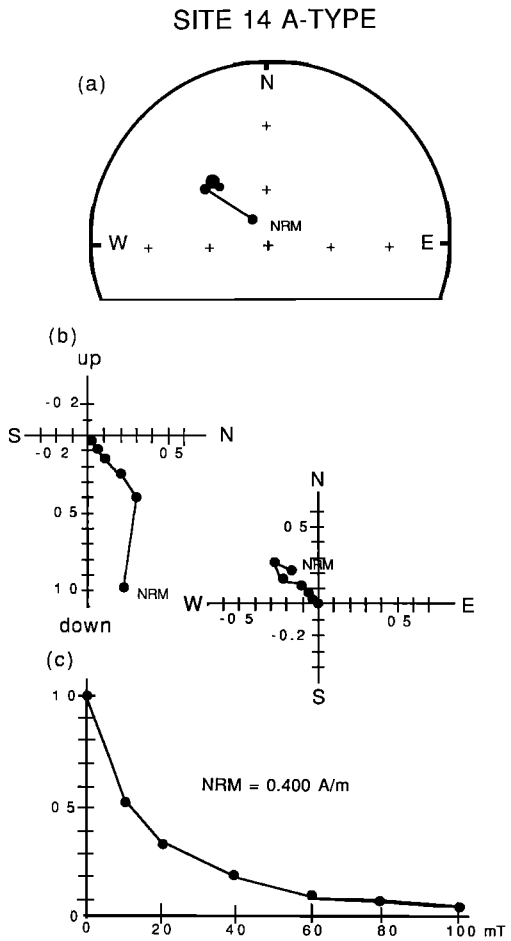


Fig. 5. A-type magnetization. (a) Equal-area projection of observed directions, (b) normalized orthogonal plots, (c) decay of intensity of magnetization during AF demagnetization. After removal of a soft magnetization, there is an excellent end-point and straight-line decay to the origin. In this and the following figures, solid (open) circles in the stereograms denote positive (negative) inclination. Directions always with respect to the present horizontal plane. NRM, natural remanent magnetization, measured in Amperes per meter (A/m).

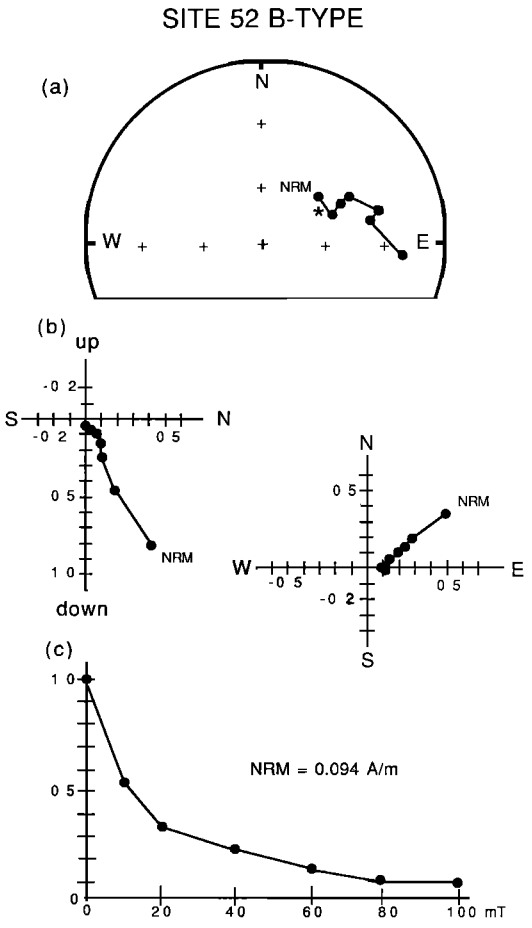


Fig. 6. B-type magnetization. (a) ,(b), and (c) as in Figure 5. The orthogonal plots show linear decay but not to the origin. There is no end-point. The star denotes the direction of magnetization removed during demagnetization in the range 20 to 100 mT.

Coryell suite of Eocene age [Little, 1957] and we refer to it as the Granby pluton because the Granby River flows across it. It is bounded to the east by the Granby River extensional fault (GF), which juxtaposes low-grade rocks to the west against high-grade rocks of the Kettle-Grand Forks metamorphic core-complex to the east (KC of Figure 4). A U/Pb zircon age of 51.1 Ma was obtained recently for the pluton by Carr and Parkinson [1989]. It should be noted that blocking temperatures for the U/Pb system are higher than observed magnetic blocking temperatures (500 to 580°C). K/Ar ages are now available on biotites extracted from dioritic rocks of the Granby pluton (Table 1). They range from 50.8 to 53.4 Ma. The closeness of the U/Pb and K/Ar dates indicates that they have cooled rather quickly. Hence, the magnetization has most probably been acquired shortly after 51 Ma.

The body is composed mainly of pink or pale gray syenite. Experience shows that rocks of this type rarely are stable magnetically. Magnetically stable, dark gray rocks of dioritic composition occur less commonly, and these have been sampled at 10 sites (Figure 10). The sites range in elevation over 500 m. Site 14 is in a dioritic dike intruded into the pluton shortly after its emplacement (J.T. Fyles, personal communication, 1988).

Site 15 has C-type magnetization. The other nine sites (54 cores, 93 specimens) have mean-site directions that, on average, are directed toward the northwest with positive inclinations (Table 2 and Figure 7). The scatter, although substantial, is random about the mean, and there is no correlation of direction with position within the pluton or with altitude.

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TABLE 1. K/Ar Ages for the "Granby" Pluton

Site	Rock	Coordinates	wt %K	Radogenic Ar (10^{-7} cc/g)	Atmospheric Ar(%)	Age (Ma±2s.d.)
15	qu.di.	3936,54608	7.26±0.3%	152.8±0.6%	4.4	53.4±0.7
18	mi.di.	3869,54800	6.02±1.3%	122.8±0.6%	3.7	51.8±1.4
21	di.	3899,54749	7.44±0.3%	148.8±0.6%	4.3	50.8±0.7

Ages obtained from R.R. Parrish, Geochronology Section, Geological Survey of Canada, Ottawa. Rock types: qu.di., quartz diorite; mi.di., micaceous diorite; di.,diorite. East and north UTM coordinates grid (zone 11U) are given. Ages determined from biotite. s.d., standard deviation from mean age.

TABLE 2. Data by Sites

Site	Rock	Coordinates	*n, s, s'*	*D°,I°*	*k*	α_{95}°	lat°N,long°E
			Granby Pluton				
13	qu.di.	3906,54504	8,15,12	318,66	64	6	62,166
14	di.	3916,54495	6,12,12	303,43	49	6	40,145
16	qu.di.	3859,54812	6,12,11	327,58	50	7	64,139
17	di.	3867,54805	8,16,16	304,27	79	4	33,134
18	mi.di.	3869,54800	5, 7, 7	308,16	119	6	30,126
19	mi.di.	3884,54795	6,12, 8	340,41	31	10	59,099
20	qu.di.	3900,54748	5,10,10	304,44	179	4	41,144
21	di.	3899,54749	5,10,10	308,40	139	4	42,137
22	di.	3954,54725	5, 9, 9	283,20	59	7	16,149
			Syringa Dikes and Associated Metamorphic Rocks				
52	di.	4284,54719	5,10,10	057,56	39	8	47,328
54	di.	4300,54699	5,10,10	055,51	104	5	45,334
56	di.	4342,54675	5,10,10	245,-54	95	5	41,324
57	di.	4363,54662	5,10,10	235,-40	201	3	40,344
58	metad.	4494,54577	5,10,10	050,42	368	3	44,347
59	metad.	4493,54577	9,14,13	038,42	79	5	51,359
01	di.	4375,54658	5,10,10	227,-34	74	6	42,354
02	di.	4375,54659	5, 9, 9	069,52	25	10	38,323
05	metad.	4420,54650	6,12,12	068,51	14	12	37,324
06	metad.	4422,54650	5,10, 9	065,37	77	6	32,337
07	metad.	4425,54648	5,10,10	052,54	20	11	50,333

Rock types as in Table 1, also metad., metamorphosed diorite. *n,* number of cores collected; *s,* number of specimens cut; *s',* number of specimens used in calculating site-mean directions. *D°,I°,* declination and inclination of the mean direction; *k,* estimate of Fisher's [1953] precision parameter; α_{95}° half-angle of cone of 95% confidence; latitude(°N) and longitude(°E) of paleopole. Data of sites 05, 06, and 07 revised from Bardoux and Irving [1989].

SYRINGA DIKES AND METADIORITES

The Syringa dikes were intruded into gneissic plutonic rocks of the Southern Valhalla Complex at a late stage in the latter's deformation (R.R. Parrish, personal communication, 1985). A few dikes have sharp contacts and are little altered. Most are deformed to varying degrees, and many of them merge almost imperceptibly into the gneiss. A U/Pb age of 51.5 ± 0.5 Ma from syenite associated with the dikes has been given by Parrish et al. [1988], and this provides an upper limit to the age of magnetization.

Nine sites, located in dikes and associated foliated

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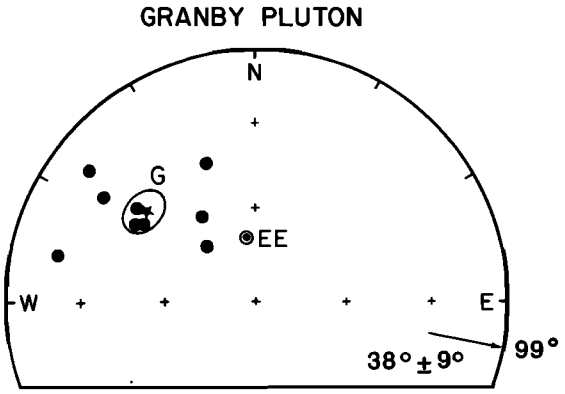


Fig. 7. Mean-site directions of the Granby Pluton. The perimeter is the present horizontal. G is the grand mean (star) and standard error circle obtained by reversing the sites with negative inclination and giving each of them unit weight (Table 3). EE is the Early Eocene paleodirection expected from the craton (Table 4). The tilt estimate is obtained by rotating G to EE about a strike line 8° east of N. Error in tilt direction is 6°.

metadiorites were sampled along Lower Arrow Lake and highway 3 (Figures 4 and 10). We will refer to this section as the "Syringa" section. Two Syringa dikes (sites 01 and 02) and three sites in metadiorite previously studied by Bardoux and Irving [1989] have been reprocessed and new data added, making a total of 14 sites.

Deformed dikes were more stable magnetically than undeformed dikes. Generally the dikes display A-type behavior (unblocking temperatures 500-580°C). Only two unaltered dikes (sites 01 and 07) yielded consistent (that is A- or B-type) data. Three dikes (sites 51, 53 and 55) had C-type magnetization and yielded no meaningful information.

The 11 site directions at the 11 remaining sites (60 cores, 113 specimens) are well-grouped, dike and metadiorite directions being in excellent agreement (Table 2 and Figure 8). Three dikes (sites 1, 56, 57) in the center of the traverse have reversed polarity. The scatter observed is random about the mean, there being no systematic variation along the "Syringa" section. The means and associated statistics are given in Table 3.

DISCUSSION

Numerous data are available from lower Eocene rocks of western North American for comparison. Data have been selected for the time interval 53 to 47 Ma within which the magnetization of the rocks just described was acquired (Table 4). There are three results from Wyoming

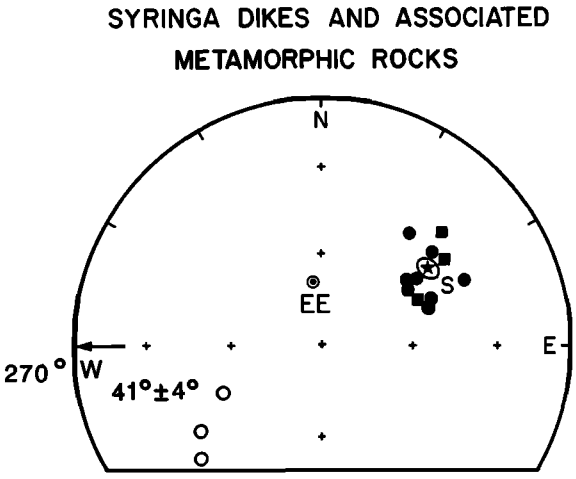


Fig. 8. Mean-site direction of Syringa dikes and associated metadiorites. The perimeter is the present horizontal. S is the grand mean (star) with standard error circle. EE as for Figure 7. Filled squares are the equivalent normal directions of the reversed polarity sites (open circles). The tilt is obtained by rotating S to EE about a strike line 4° west of north. Error in tilt direction is 2°.

TABLE 3. Summary of Data

	N	$D°,I°$	k	$\alpha_{63}°$	$\alpha_{95}°$	lat°N,long°E
			Granby Pluton			
Directions	9	309,41	16	7	14	42,137
Site pole	9	-----	17	7	13	44,138
			Syringa Dikes and Associated Metamorphic Rocks			
Directions	11	56,47	63	3	6	56,318
Site poles	11	-----	58	3	6	56,319
			Expected Values			
Paleopoles	5	-----	549	2	3	83,190
Directions	5	351,67	---	1	2	------

N number of sampling sites. $D°$, $I°$, k, $\alpha_{63}°$, $\alpha_{95}°$, lat°N, long°E as in Table 2.

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TABLE 4. Paleopoles From Coeval Rocks in Adjacent Regions

Note	Rock Unit	age (Ma)	N	lat°N,long°E	A_{95}°
(1)	Bearpaw Mountains volcanics, Montana	50-54	30	81,189	6
(2)	Absaroka Supergroup volcanics, Wyoming	48	19	84,177	10
(3)	Highwood Mountains volcanics, Montana	49-53	30	82,163	7
(4)	Mean of 1, 2 and 3 (C of Fig. 9)		3	83,176	3
(5)	Kelowna volcanics, British Columbia	52	28	85,197	10
(6)	Kamloops volcanics, British Columbia	49	26	81,222	12
(7)	Flores volcanics, British Columbia	55	12	81,188	10
(8)	Mean of 1,2,3,5,6,7		6	83,190	3

(1) and (3) Diehl et al. [1983]. (2) Shive and Pruss [1977]. (5) Bardoux and Irving [1989]. (6) Symons and Welling [1989]. (7) Irving and Brandon [1990]. N number of sampling sites. The mean entry (8) is of very high precision, and the full statistics are lat 82.7°N, long 189.9°E, K = 638, A_{63}° = 1.4°, A_{95}° = 2.7°. Paleofield direction expected at center of the sampling region (45.3°N 118.1°W) is (D°, I°) = 351.1°, 66.9°; α_{95}° = 2.0°.

and Montana, and three from the western Cordillera of Canada. Their paleopoles are in excellent agreement. This means that in the time interval during which the rocks of the Granby pluton and the "Syringa" section acquired their magnetization, the geomagnetic field remained fixed in orientation relative to the sampling region. The mean paleopole provides an accurate measure of the reference paleofield for this region for this interval of time. Our sampling region is straddled by the localities from which these results have been obtained (Figure 9). Hence, the paleofield direction expected at both sampling localities can be accurately calculated and is the same to the nearest degree, - (D°,I°) 352°,67°, α_{95}° = 2°.

The two localities (G and S) from the zone of Eocene extension are situated between those from which the cratonic and Cordilleran data were taken, yet their paleopoles differ by amounts far exceeding experimental error. The excellent agreement among data from the Cordillera (Figure 9) and the craton shows that these regions were in their present relative positions by early Eocene time. Hence, the aberrance of paleopoles G and S cannot be explained by large tectonic translations. We conclude that they were caused by tilting, - tilting that occurred during the extensional process.

The tilt required to explain the aberrancies are down 38° ± 8° at 99°E for the Granby pluton, and for the Syringa dikes and associated metadiorites down 41 ± 4° at 270°W (Figures 7 and 8). The consistency of the data through the Syringa section does not necessarily mean that the section tilted en bloc. It could equally well reflect the similarity of dips among smaller blocks. These tilts, together with determinations from a small Coryell Syenite stock at Ymir [Bardoux and Irving, 1989] and the mid-

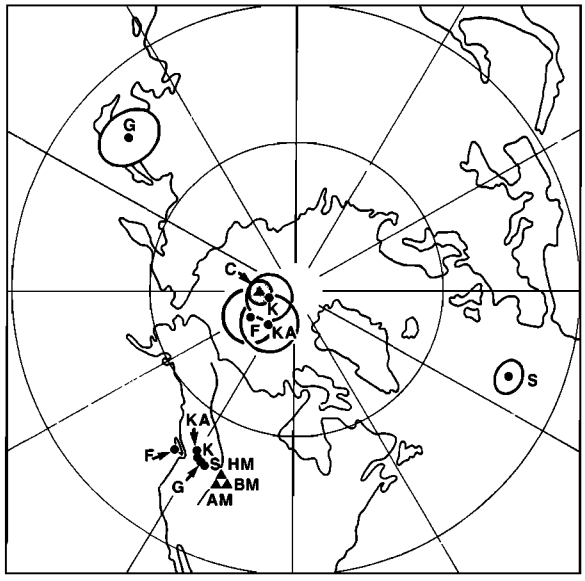


Fig. 9. Paleopoles from Lower Eocene rocks (48 to 55 Ma) from the craton (HM, BM, AM) and from the Canadian Cordillera compared. Standard error circles are drawn. C is the cratonic reference paleopole. G and S are paleopoles from the Granby pluton and the Syringa dikes and metadiorite, HM, BM, AM are paleopoles from the Highwood and Bearpaw Mountains (Montana) and from the Absaroka Supergroup (Wyoming). K, KA and F are paleopoles obtained from the Kelowna and Kamloops volcanics and from the Flores volcanics of western Vancouver Island. Documentation in Table 4.

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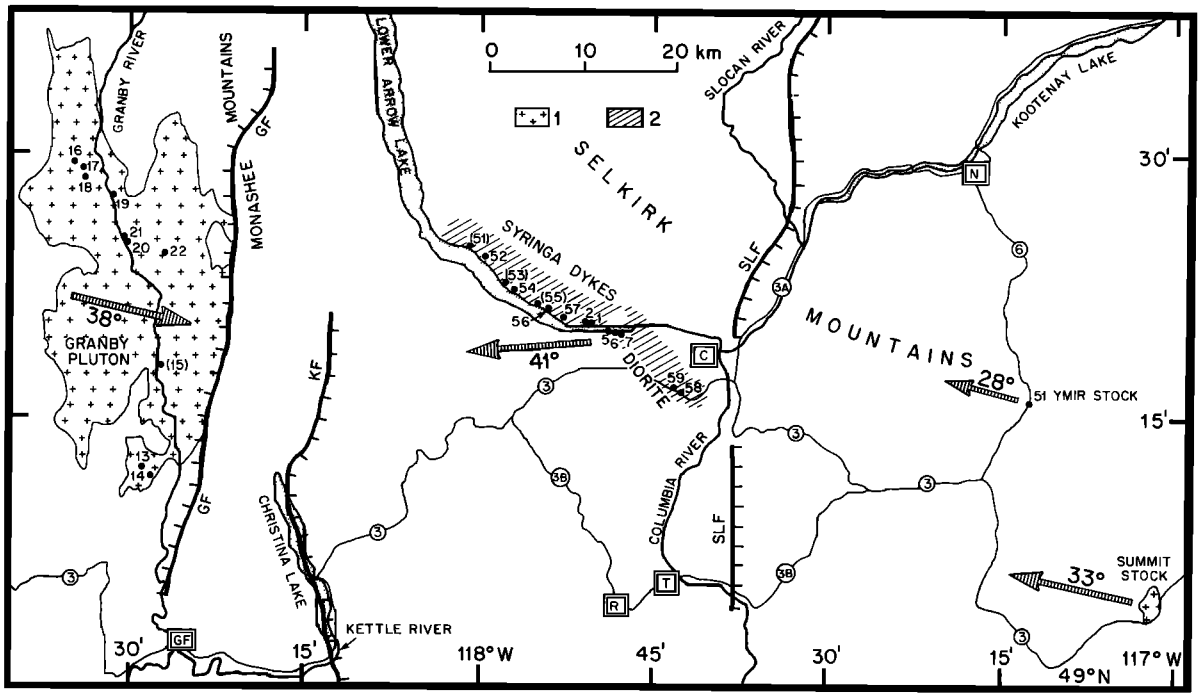


Fig. 10. Sketch-map of sampling region compiled mainly from Parrish et al. [1988]. The sampling sites are numbered as in Table 1. Sites in brackets yielded C-type magnetization only. The towns of Grand Forks (GF), Rossland (R), Trail (T), Castlegar (C) and Nelson (N) are shown. Main highways are numbered. Large arrows give the paleomagnetic estimates of tilt. Two other estimates, from the Eocene Ymir stock [Bardoux and Irving, 1989] and for the Cretaceous Summit stock [Irving and Archibald, 1990] are also given.

Cretaceous Summit Stock [Irving and Archibald, 1990] have been compiled in Figure 10. The tilts range from 28° to 41°. The tilts are to the west when situated to the east of, and to the east when west of the central culmination, the Kettle-Grand Forks core-complex (Figures 4 and 10). The sense of tilt is always opposite to that of the underlying extensional faults, the west-dipping Granby River fault and the east-dipping Kettle River fault.

The tilts observed are consistent with the antithetic hanging-wall fault-block rotations predicted by the Wernicke-Buck model (Figure 2). Such tilts would have been produced if the Granby, Kettle River and Slocan Lake faults of Figure 4 are the surviving lower parts of listric normal faults which failed sequentially one-by-one from east to west.

Although Parrish et al. [1988] do not discuss back-tilting in their paper, back-tilting may be considered to be implied by their model. Firstly their model may be considered as a variation of a simple "card-deck" type of extension, as described by Davis [1983]. This model predicts tilts in the directions we have observed, but the amount of extension required to obtain tilts of the order of 40° is in excess of 200%. Parrish et al. [1988] conclude that the amount of extension in this region is about 25%. If they are correct then back-tilting of the magnitude we observe cannot be explained by card-deck extension in this region.

Secondly Parrish et al. [1988] show that the trace of the Valkyr Shear Zone (an older system which was cut by the Slocan Lake fault), between the Kettle River and Slocan Lake faults, is curved concave upwards. If the Valkyr Shear Zone were once planar (for which there seems to be no evidence) then its curved nature implies back-tilting between the Kettle River and Slocan Lake faults as we have observed in the Syringa section. However the back-tilting implied would vary widely from 20°W just west of the Slocan Lake fault to 50°W as the Kettle River fault is approached. This contrasts with the uniform and systematic nature of the tilts observed paleomagnetically. Thirdly the traces of the throughgoing Late Cretaceous faults displayed in the sections of Parrish et al. [1988] show post-50 Ma tilting by amounts that are variable but which rarely exceed 10°, again inconsistent with the larger tilts presented in this study. We therefore conclude that these tilt relationships are not easily reconciled with the extensional fault model of Parrish et al. [1988].

This study shows that, under certain circumstances, tilts can be accurately estimated paleomagnetically in unbedded rocks of the middle crust. Because the reference paleofield for the early-middle Eocene is known very accurately, and because any large scale translations and rotations that the region might have undergone had been completed by this time, the paleomagnetic method is capable of supplying this fundamental structural information. There seems to be

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no reason why tilts of plutonic and metamorphic core-complex rocks cannot be determined throughout the region. In addition to restoring Eocene structures, such results may shed light on the interpretation of older Mesozoic structures, whose geometrical relationships depend on correctly removing Eocene strain.

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